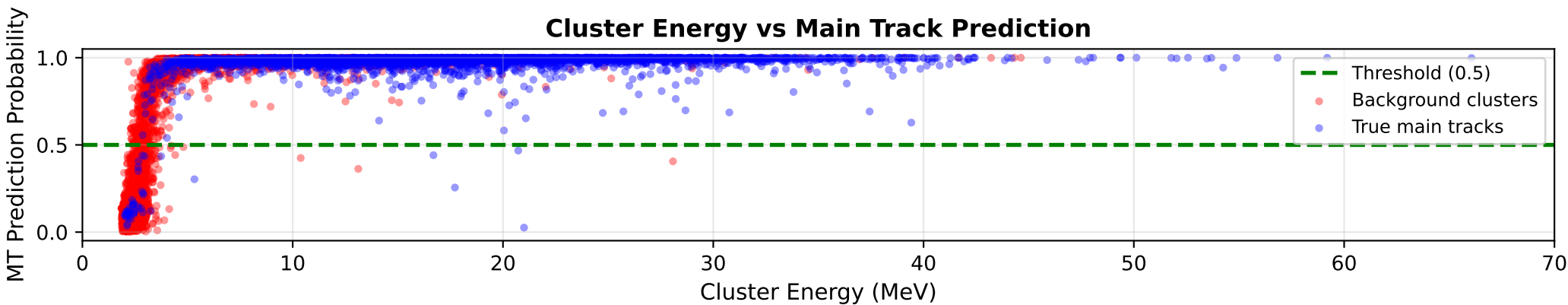


MT Prediction Analysis - Energy vs Prediction



MT INFERENCE SUMMARY - CAT000001

=====

DATASET:

Total clusters:	19,384
True main tracks:	3,789
True backgrounds:	15,595

PREDICTIONS:

Predicted main tracks:	6,814
Predicted backgrounds:	12,570

CONFUSION MATRIX:

True Positives (TP):	3,742
False Positives (FP):	3,072
False Negatives (FN):	47
True Negatives (TN):	12,523

PERFORMANCE METRICS:

Accuracy:	83.9%
Precision:	54.9%
Recall (Efficiency):	98.8%
Contamination (FP rate):	45.1%

ENERGY ANALYSIS:

True Positives:

Mean:	17.93 MeV
Median:	16.91 MeV
Range:	2.87 - 66.06 MeV

False Positives:

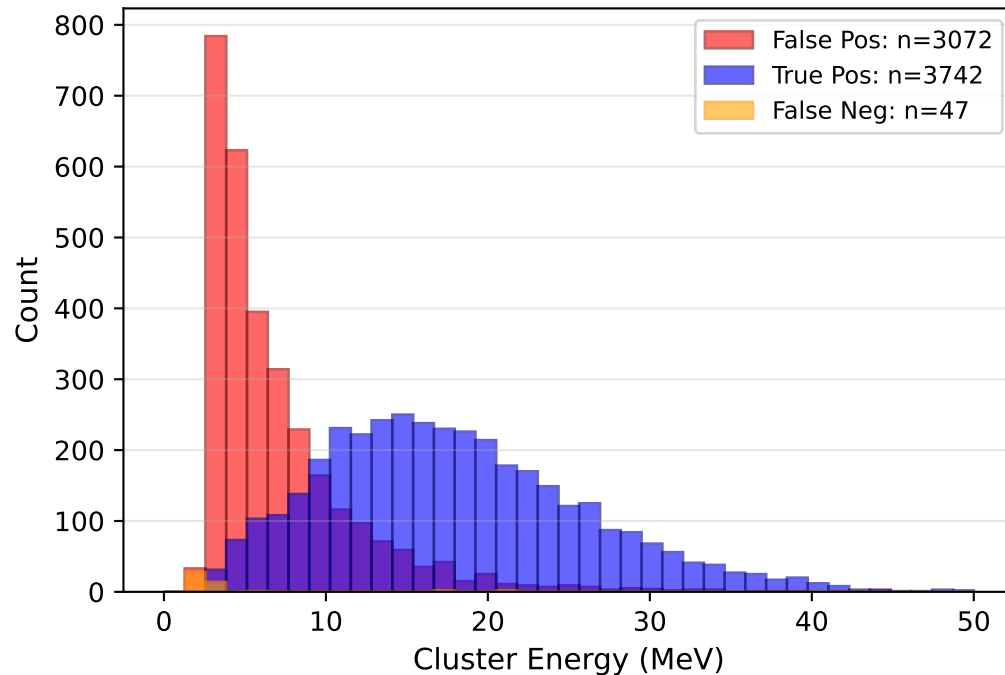
Mean:	7.03 MeV
Median:	5.44 MeV
Range:	2.18 - 44.63 MeV

LOW ENERGY ANALYSIS (< 5 MeV):

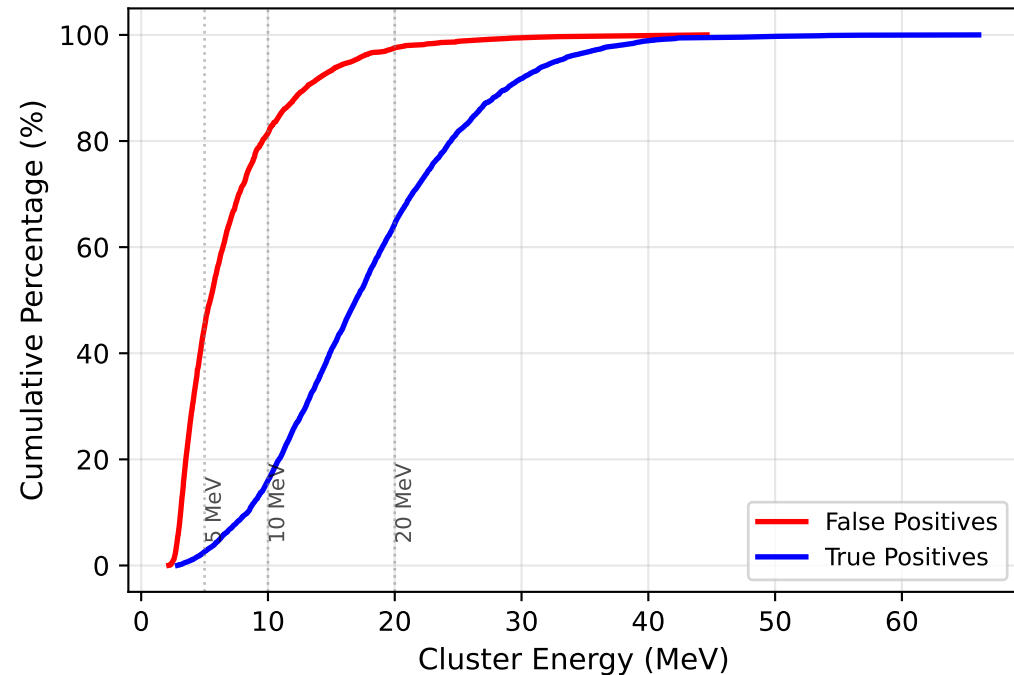
FP below 5 MeV:	1,385 (45.1% of all FP)
TP below 5 MeV:	98 (2.6% of all TP)

MT Prediction Analysis - Detailed Distributions

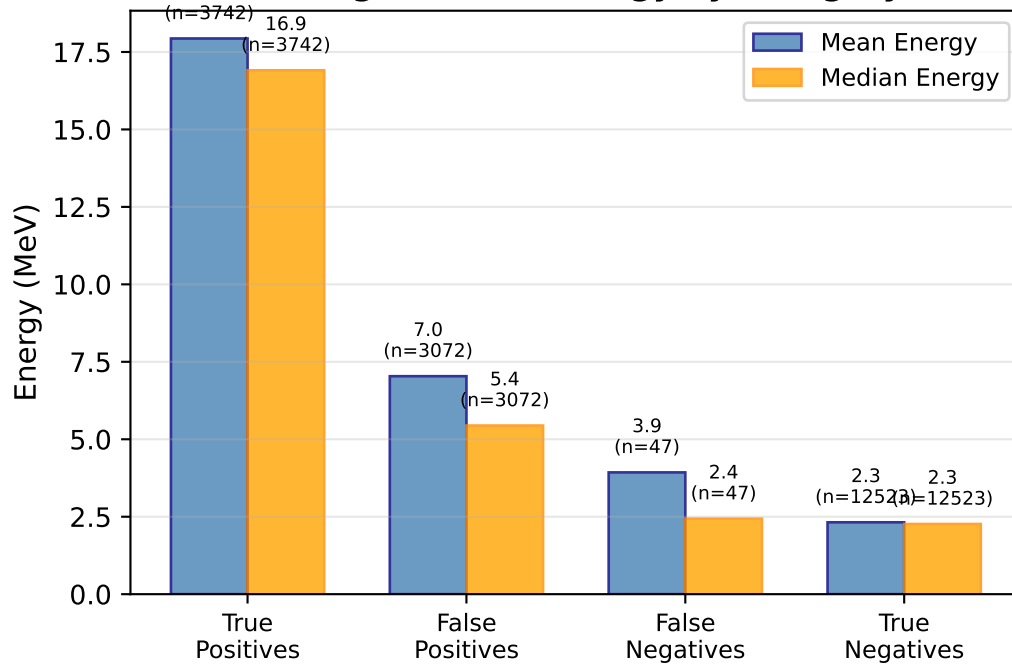
Energy Distribution by Prediction Outcome



Cumulative Energy Distribution



Average Cluster Energy by Category



Distribution of MT Predictions

